

Midterm Project: Third-Person Combat Simulation

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Course: 3D Game Applications

Engine Version: Unreal Engine 5 (Spring 2026)

1. Project Overview

This project is a third-person combat simulation built using the Unreal Engine 5 Third Person template. Following the assignment constraints, the entire environment and all characters are constructed using primitive meshes (cubes, spheres, and cylinders). The game features a player-controlled character facing off against three AI-driven enemies in a tactical arena.

2. Core Features

- **Combat System:** Both player and AI use a projectile-based system. Each actor starts with 10 HP, and each hit reduces health by 1 HP.
- **Custom HUD:** A Widget Blueprint displays the player's health bar and remaining ammunition.
- **AI Behavior Tree:** Enemies use a structured Behavior Tree to make decisions:
 - **Retreat:** Maintain a tactical distance if the player is too close.
 - **Attack:** Fire projectiles when the player is within the Line of Sight (LOS).
 - **Roam:** Move to random locations within the arena when the player is not detected.
- **Win/Loss Conditions:** * **Victory:** Eliminate all 3 enemies.
 - **Defeat:** Reaching 0 HP or running out of ammunition.

3. Added Feature (Bonus)

- **Independent Enemy Ammo Tracking:** While not strictly required by the initial prompt, I implemented an individual ammunition system for each AI.
 - Each enemy has a limit of **30 projectiles**.
 - Once an enemy's ammo is depleted, they stop firing and switch permanently to their "Roam" (patrol) behavior, making the combat more dynamic.

4. Technical Implementation

- **Navigation:** A `NavMesh Bounds Volume` is used for AI pathfinding.
- **Sensing:** A custom `BTService` updates the player's location and visibility status every 0.3 seconds.
- **Environment:** The arena was designed using BSP brushes and primitives, strategically placed to offer cover and break the AI's Line of Sight.

5. Controls

- **Move:** W, A, S, D / Z, Q, S, D
- **Jump:** Space Bar
- **Fire:** Left Mouse Button
- **Restart:** Click the "Restart" button on the Game Over/Victory screen.